APPROXIMATION GLUES ACROSS AN ELEMENTARY DISTINGUISHED SQUARE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3640-3648, immediately following the statement at 3627-3638.

Editorial note. Hypothesis (2) (cohomology sheaves supported on T) enters through the stalk equation reading of j_!P's support.

Allowed standard prerequisites (editorial).

- Pushforward with support in an open. Lemma lemma-pushforward-with-support-in-open of this chapter, cited at proof lines 3643 and 3647. Rj_*P restricted to V equals P; and Rj_*P = j_!P, the extension-by-zero identification.
- Perfectness of the supported pushforward. Lemma lemma-pushforward-perfect of this chapter, cited at proof lines 3641-3642. Rj_*P is perfect on X when P is a perfect approximation over V.
- Stalks of extension by zero. equation-stalk-j-shriek of this chapter, cited at proof lines 3645-3646. Reads off that j_!P is supported on T.

Source reference. Original definition 08HI: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 3575–3575. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original equation 08GU: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1810–1810. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 08HK: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 3610–3610. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 08GG: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 2135–2135. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let $(U \subset X, j : V \to X)$ be an elementary distinguished square of algebraic spaces over S. Let T be a closed subset of $|X| \setminus |U|$ and let (T, E, m) be a triple as in Definition 08HI. If*

(1) *approximation holds for $(j^{-1}T, E|_V, m)$, and*
(2) *the sheaves $H^i(E)$ for $i \geq m$ are supported on T,*

then approximation holds for (T, E, m).

Proof. Let $P \to E|_V$ be an approximation of the triple $(j^{-1}T, E|_V, m)$ over V. Then Rj_*P is a perfect object of $D(\mathcal{O}_X)$ by Lemma 08HK. On the other hand, $Rj_*P = j_!P$ by Lemma 08GG. We see that $j_!P$ is supported on T for example by (08GU). Hence we obtain an approximation $Rj_*P = j_!P \to j_!(E|_V) \to E$. □